

EGRE 245 Engineering Programming**Fall 2025**

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Programming Assignment 5: Hospital Patient Tracker

1 Overview

In this assignment, you will write a C program to manage patient records for a small hospital intake system. Patient data will be stored in an external text file, and your program will read the data into dynamically allocated memory. You will then use pointers and structures to analyze, display, and update the data.

This assignment reinforces:

- File input using `fopen()`, `fscanf()`, and `fclose()`
- Structures and arrays of structures
- Pointer-to-structure notation (`-i`)
- Returning pointers from functions
- Dynamic memory allocation using `malloc()` and `free()`

2 Patient Structure

Each patient record has the following fields:

```
struct Patient {  
    char name[30];  
    int id;  
    int age;  
    float temp;  
    int condition;    // 1 = stable, 2 = observation, 3 = critical  
};
```

3 Input File Format

The input file `patients.txt` will contain one patient per line, with the following format:

Name ID Age Temperature Condition

Example:

```
Alice_Smith 101 42 98.7 1
Bob_Jones   102 67 101.2 2
Carol_White 103 55 99.1 1
```

Note: Spaces in names are replaced with underscores.

4 Program Requirements

Your program must perform the following tasks:

1. Ask the user for the number of patients in the file.
2. Allocate memory dynamically:
3. Open `patients.txt` and read exactly `n` records into the array.
4. Write and use the following pointer-based functions:
 - `void displayPatient(const struct Patient *p)`
 - `float averageTemp(struct Patient *list, int n)`
 - `struct Patient *findOldest(struct Patient *list, int n)`
 - `void promoteCondition(struct Patient *p)`
5. Display all patient data in a table.
6. Compute and display the average temperature.
7. Find and display the oldest patient.
8. Ask the user for a patient ID and promote that patient's condition (1→2→3, but do not go beyond 3).
9. Display the updated list.
10. Free all dynamically allocated memory.

5 Sample Output

Your output should match the sample run below.

```
Enter number of patients in file: 5
Reading data from file...

-----
Name           ID      Age      Temp      Cond
-----
Alice_Smith    101     42      98.70     1
Bob_Jones      102     67      101.20    2
Carol_White    103     55      99.10     1
David_Clark    104     73      100.40    3
Eva_Brown      105     36      98.50     1

Average temperature = 99.58
Oldest patient: David_Clark (age 73)

Enter ID to promote condition: 105
Updated condition for Eva_Brown = 2

Updated Patient List:
-----
Name           ID      Age      Temp      Cond
-----
Alice_Smith    101     42      98.70     1
Bob_Jones      102     67      101.20    2
Carol_White    103     55      99.10     1
David_Clark    104     73      100.40    3
Eva_Brown      105     36      98.50     2

...Program finished with exit code 0
Press ENTER to exit console.[]
```

6 Deliverables

Submit a single .c file named:

Assignment_PointersStructs_FirstName_LastName.c

Your submission must include:

- Proper header comment block
- Function prototypes
- Internal documentation and clear formatting

This assignment is due Saturday, November 22, at 11:59 pm.